

Pinky 주피터로 내비게이션 하기



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Pinky 주피터로 내비게이션 하기

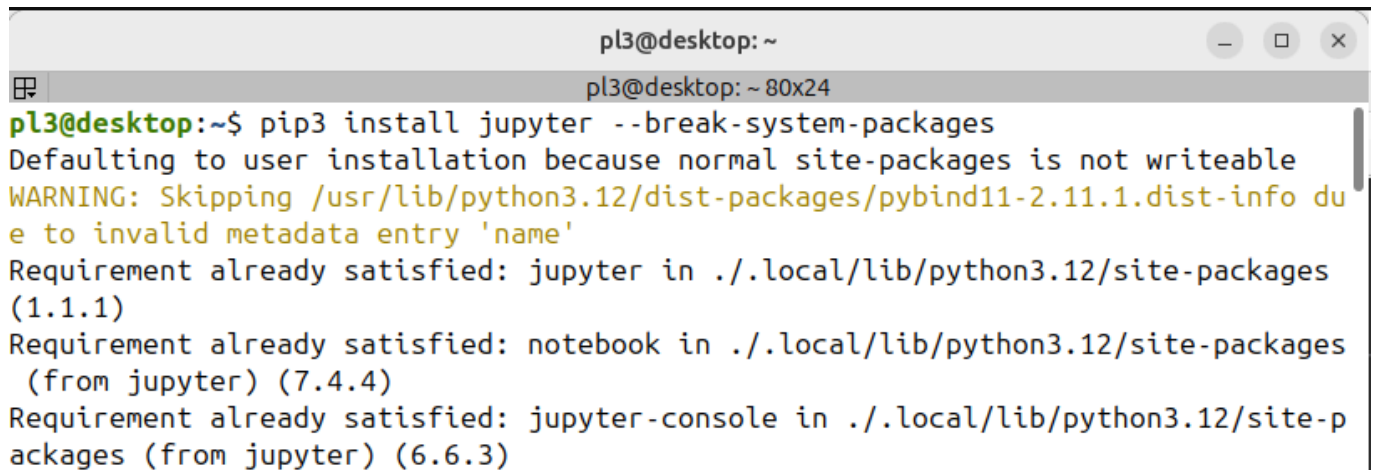
Exported from Confluence · <https://pinkwink.atlassian.net>

주피터 설치

최신 파이썬 정책이 바뀌어서([PEP668](https://peps.python.org/pep-0668/)) 가상환경을 사용하도록 권장해서 전역 패키지로 설치하려면 `--break-system-packages` 옵션을 추가해야한다

Code

```
1 pip3 install jupyter --break-system-packages
```



```
pl3@desktop: ~  
pl3@desktop: ~ 80x24  
pl3@desktop:~$ pip3 install jupyter --break-system-packages  
Defaulting to user installation because normal site-packages is not writeable  
WARNING: Skipping /usr/lib/python3.12/dist-packages/pybind11-2.11.1.dist-info due to invalid metadata entry 'name'  
Requirement already satisfied: jupyter in ~/.local/lib/python3.12/site-packages (1.1.1)  
Requirement already satisfied: notebook in ~/.local/lib/python3.12/site-packages (from jupyter) (7.4.4)  
Requirement already satisfied: jupyter-console in ~/.local/lib/python3.12/site-packages (from jupyter) (6.6.3)
```

[로봇] pinky bringup 실행

Code

```
1 ros2 launch pinky_bringup bringup_robot.launch.xml  
2 ros2 launch vicpinky_bringup bringup.launch.xml
```

```
pinky@raspi:~$ ros2 launch pinky_bringup bringup_robot.launch.xml
[INFO] [launch]: All log files can be found below /home/pinky/.ros
/log/2025-10-20-14-01-42-797058-raspi-5276
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [5280]
[INFO] [joint_state_publisher-2]: process started with pid [5281]
[INFO] [sllidar_node-3]: process started with pid [5282]
[INFO] [bringup-4]: process started with pid [5283]
[sllidar_node-3] [INFO] [1760936503.348025118] [sllidar_node]: SLL
idar running on ROS2 package SLLidar.ROS2 SDK Version:1.0.1, SLLID
AR SDK Version:2.1.0
[robot_state_publisher-1] [INFO] [1760936503.365403449] [robot_sta
te_publisher]: Robot initialized
```

[로봇] 내비게이션 실행

Code

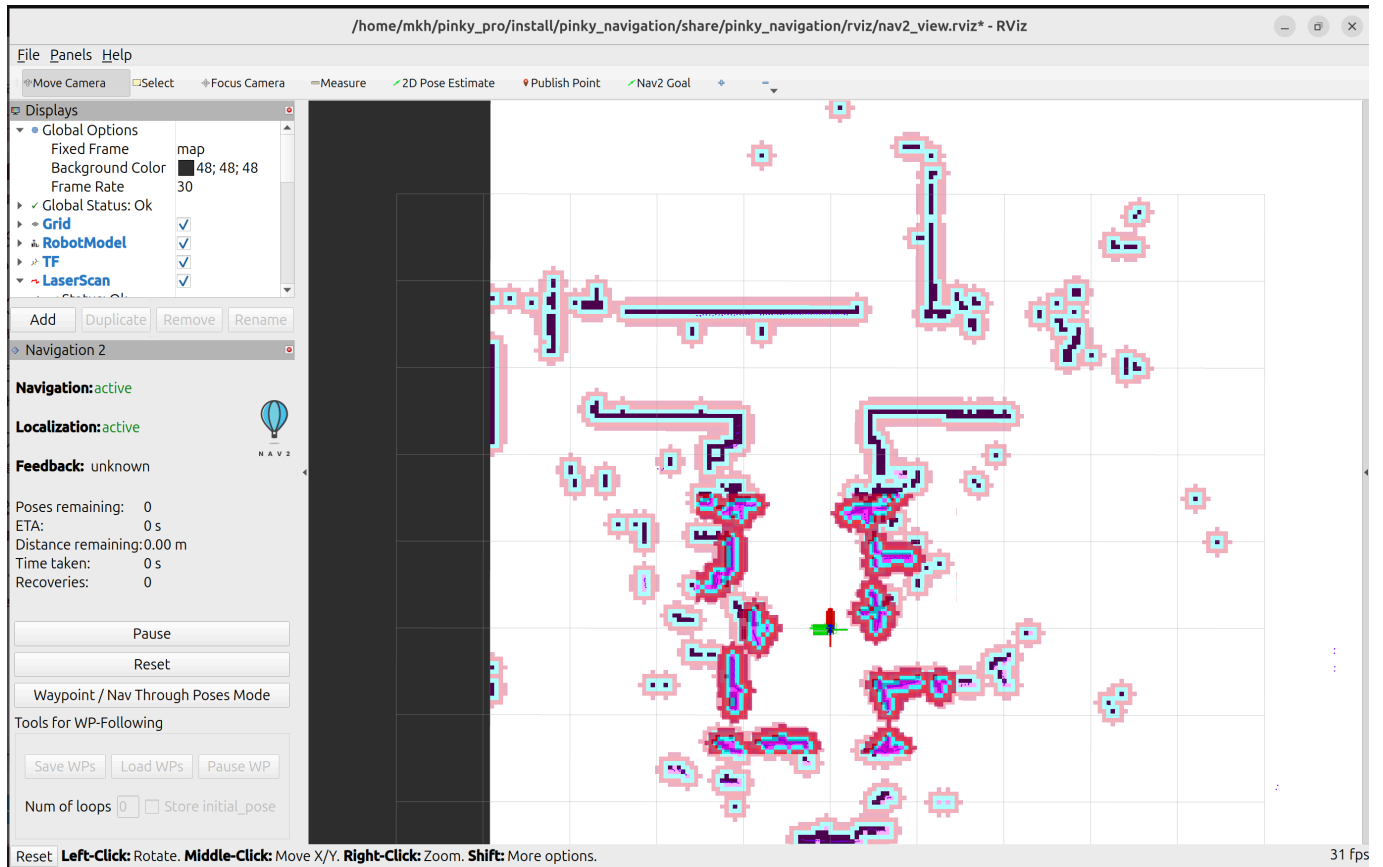
```
1 ros2 launch pinky_navigation bringup_launch.xml map:=<저장한 맵이름.yaml>
2 ros2 launch vicpinky_navigation bringup_launch.xml map:=저장한_맵.yaml
```

```
pinky@raspi:~$ ros2 launch pinky_navigation bringup_launch.xml map:=my_map
.yaml
[INFO] [launch]: All log files can be found below /home/pinky/.ros/log/202
5-01-13-23-41-20-517658-raspi-2974
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [component_container_isolated-1]: process started with pid [2995]
[INFO] [cartographer_node-2]: process started with pid [2996]
[component_container_isolated-1] [INFO] [1736779281.944944101] [nav2_conta
iner]: Load Library: /opt/ros/jazzy/lib/libmap_server_core.so
[cartographer_node-2] [INFO] [1736779281.976917743] [cartographer logger]:
I20250113 23:41:21.-2147483648 2996 configuration_file_resolver.cc:41] F
ound '/home/pinky/pinky_violet/install/pinky_cartographer/share/pinky_cart
ographer/params/nav2_cartographer_params.lua' for 'nav2_cartographer_param
s.lua'.
[cartographer_node-2] [INFO] [1736779281.977852329] [cartographer logger]:
I20250113 23:41:21.-2147483648 2996 configuration_file_resolver.cc:41] F
ound '/opt/ros/jazzy/share/cartographer/configuration_files/map_builder.lu
a' for 'map_builder.lua'.
[cartographer_node-2] [INFO] [1736779281.977908700] [cartographer logger]:
I20250113 23:41:21.-2147483648 2996 configuration_file_resolver.cc:41] F
ound '/opt/ros/jazzy/share/cartographer/configuration_files/map_builder.lu
a' for 'map_builder.lua'.
```

[PC] nav2_view.rviz 실행

Code

```
1 ros2 launch pinky_navigation nav2_view.launch.xml
2 ros2 launch vicpinky_navigation nav2_view.launch.xml
```



2D Pose Estimate 버튼을 눌러 라이다와 맵 일치 시켜주기



주피터 실행

주피터 실행 전에 ROS2 환경이 설정되어 있는지 확인

Code

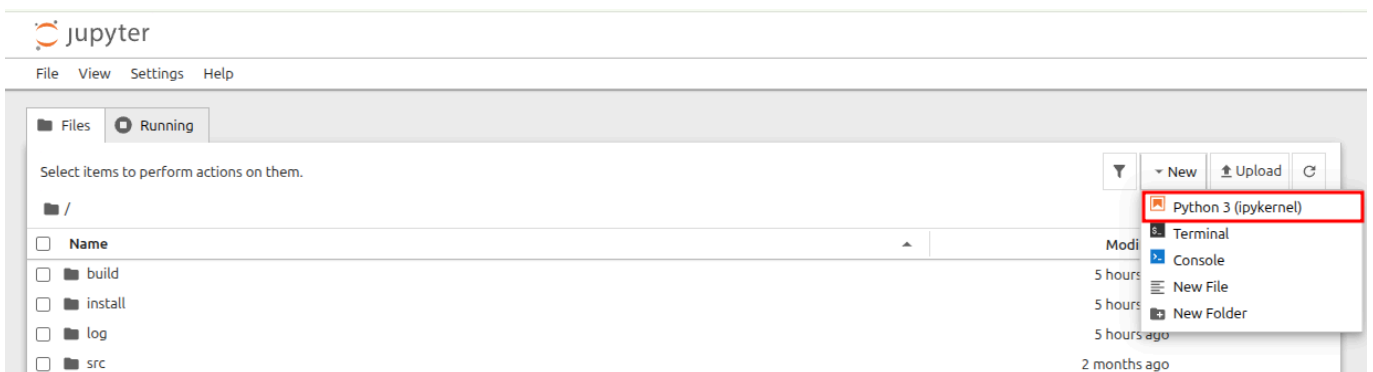
```
1 jupyter notebook
```

```

pl3@desktop:~/pinky_ws$ jupyter notebook
[I 2025-07-22 18:32:05.009 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2025-07-22 18:32:05.010 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2025-07-22 18:32:05.012 ServerApp] jupyterlab | extension was successfully linked.
[I 2025-07-22 18:32:05.014 ServerApp] notebook | extension was successfully linked.
[I 2025-07-22 18:32:05.135 ServerApp] notebook_shim | extension was successfully linked.
[I 2025-07-22 18:32:05.141 ServerApp] notebook_shim | extension was successfully loaded.
[I 2025-07-22 18:32:05.142 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2025-07-22 18:32:05.143 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2025-07-22 18:32:05.144 LabApp] JupyterLab extension loaded from /home/pl3/.local/lib/python3.12/site-packages/jupyterlab
[I 2025-07-22 18:32:05.144 LabApp] JupyterLab application directory is /home/pl3/.local/share/jupyter/lab
[I 2025-07-22 18:32:05.144 LabApp] Extension Manager is 'pypi'.
[I 2025-07-22 18:32:05.166 ServerApp] jupyterlab | extension was successfully loaded.
[I 2025-07-22 18:32:05.168 ServerApp] notebook | extension was successfully loaded.
[I 2025-07-22 18:32:05.168 ServerApp] Serving notebooks from local directory: /home/pl3/pinky_ws
[I 2025-07-22 18:32:05.168 ServerApp] Jupyter Server 2.16.0 is running at:
[I 2025-07-22 18:32:05.168 ServerApp] http://localhost:8888/tree?token=619211feefd07cb626c6

```

주피터 화면에서 new를 눌러 새 주피터 파일 생성



Nav2 클래스 불러오기

```

[1]: from nav2_simple_commander.robot_navigator import BasicNavigator
import rclpy

rclpy.init()
nav = BasicNavigator()

```

Nav2가 실행 되었는지 확인

```

[2]: nav.waitUntilNav2Active()

```

[INFO] [1753177309.048487664] [basic_navigator]: Nav2 is ready for use!

로봇의 각도를(degree) 쿼터니언으로 변환하는 함수 선언


```
[3]: import math
import tf_transformations

def get_quaternion_from_yaw(yaw_degrees):
    yaw_radians = math.radians(yaw_degrees)

    quaternion = tf_transformations.quaternion_from_euler(0, 0, yaw_radians)

    return quaternion
```

현재 위치를 출력하기 위해 amcl_pose 토픽을 구독하는 노드 하나 만들기

```
[5]: sub_amcl = rclpy.create_node('sub_amcl')

[6]: def amcl_callback(msg):
    print("===")
    print(msg.pose.pose.position)
    print(msg.pose.pose.orientation)

[7]: from geometry_msgs.msg import PoseWithCovarianceStamped

sub_amcl.create_subscription(PoseWithCovarianceStamped, 'amcl_pose', amcl_callback, 10)

[7]: <rclpy.subscription.Subscription at 0x736401df2750>
```

제자리에서 회전하는 goal 생성하기

- x, y는 로봇의 현재위치로 수정하기
- 로봇의 현재위치는 clicked_point 토픽 확인

```
*[8]: from geometry_msgs.msg import PoseStamped

goal_yaw = 180
q = get_quaternion_from_yaw(goal_yaw)

goal_pose = PoseStamped()
goal_pose.header.frame_id = 'map'
goal_pose.header.stamp = nav.get_clock().now().to_msg()
goal_pose.pose.position.x = 0.5
goal_pose.pose.position.y = 0.0
goal_pose.pose.position.z = 0.0
goal_pose.pose.orientation.x = q[0]
goal_pose.pose.orientation.y = q[1]
goal_pose.pose.orientation.z = q[2]
goal_pose.pose.orientation.w = q[3]
```

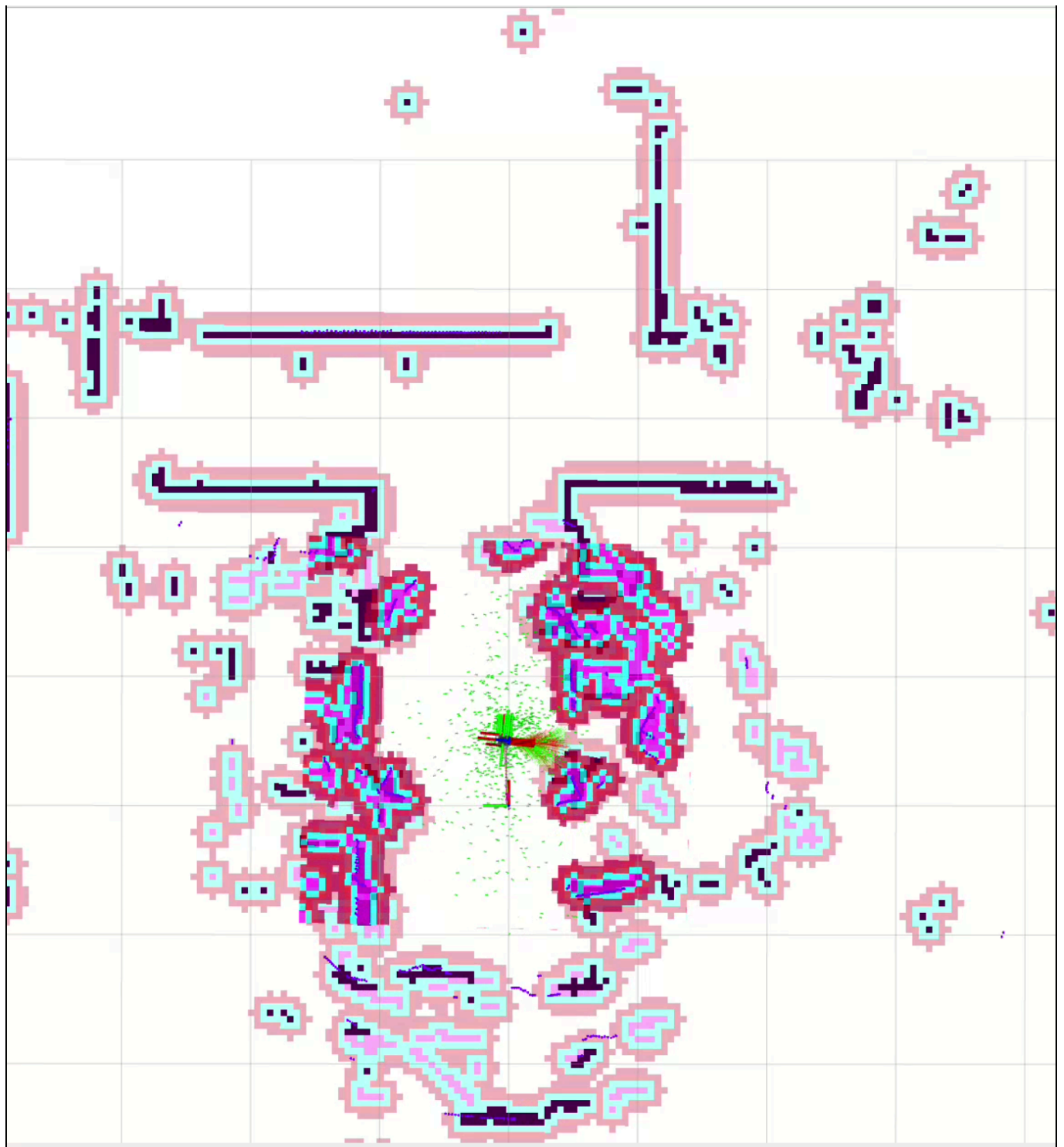
nav2에 goal 좌표를 보내면

```
[9]: nav.goToPose(goal_pose)

[INFO] [1753177309.110588891] [basic_navigator]: Navigating to goal: 0.0 0.0...

[9]: True
```

로봇이 제자리에서 회전한다



이번엔 이동하기 위해 goal 좌표 생성 후 goal 보내면

```

•[10]: from geometry_msgs.msg import PoseStamped

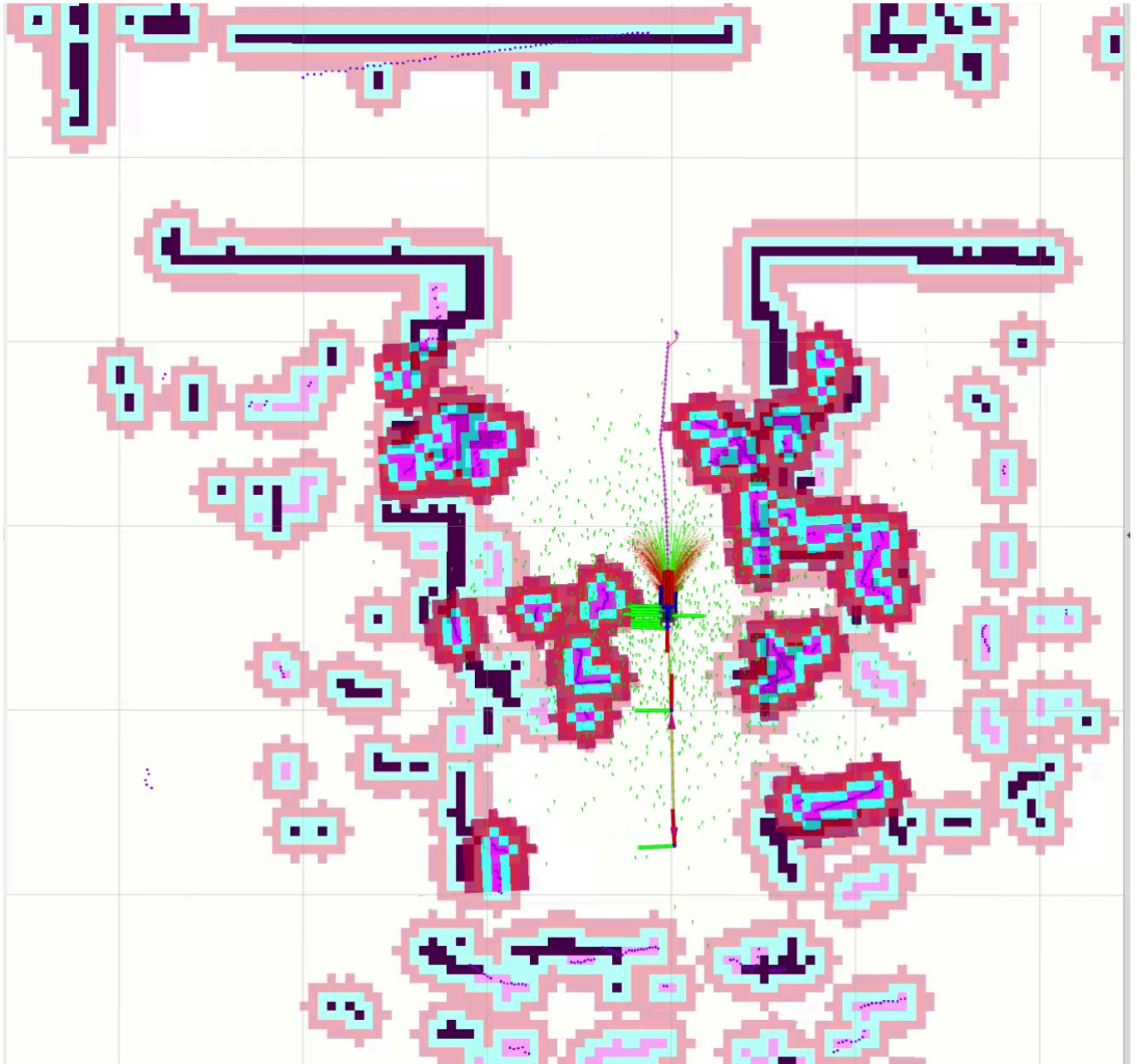
goal_yaw = 90
q = get_quaternion_from_yaw(goal_yaw)

goal_pose = PoseStamped()
goal_pose.header.frame_id = 'map'
goal_pose.header.stamp = nav.get_clock().now().to_msg()
goal_pose.pose.position.x = 2.0
goal_pose.pose.position.y = 0.0
goal_pose.pose.position.z = 0.0
goal_pose.pose.orientation.x = q[0]
goal_pose.pose.orientation.y = q[1]
goal_pose.pose.orientation.z = q[2]
goal_pose.pose.orientation.w = q[3]

[11]: nav.goToPose(goal_pose)

```

그 좌표로 이동하기 시작한다



이때 다음 코드로 피드백 출력 해보기

- 주행중에 실행하기
- cpu 부하방지 (jupyter 멈춤방지)를 위해 timeout_sec 1초로 설정

```
[*]: from rclpy.duration import Duration
from IPython.display import clear_output

while not nav.isTaskComplete():
    clear_output(wait=True)
    rclpy.spin_once(sub_amcl, timeout_sec=1.0)

    feedback = nav.getFeedback()
    print("===")
    print('Distance remaining: ' + '{:.2f}'.format(
        feedback.distance_remaining) + 'meters.')

    if Duration.from_msg(feedback.navigation_time) > Duration(seconds=30.0):
        nav.cancelTask()

===
geometry_msgs.msg.Point(x=0.14311876822926145, y=0.6280523547879696, z=0.0)
geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.7468798476231279, w=0.6649590161915645)
===
Distance remaining: 0.54meters.
```

주행 결과 확인

```
[32]: from nav2_simple_commander.robot_navigator import TaskResult

result=nav.getResult()
if result == TaskResult.SUCCEEDED:
    print('Goal succeeded!')
elif result == TaskResult.CANCELED:
    print('Goal was canceled!')
elif result == TaskResult.FAILED:
    print('Goal failed!')
else:
    print('Goal has an invalid return status!')

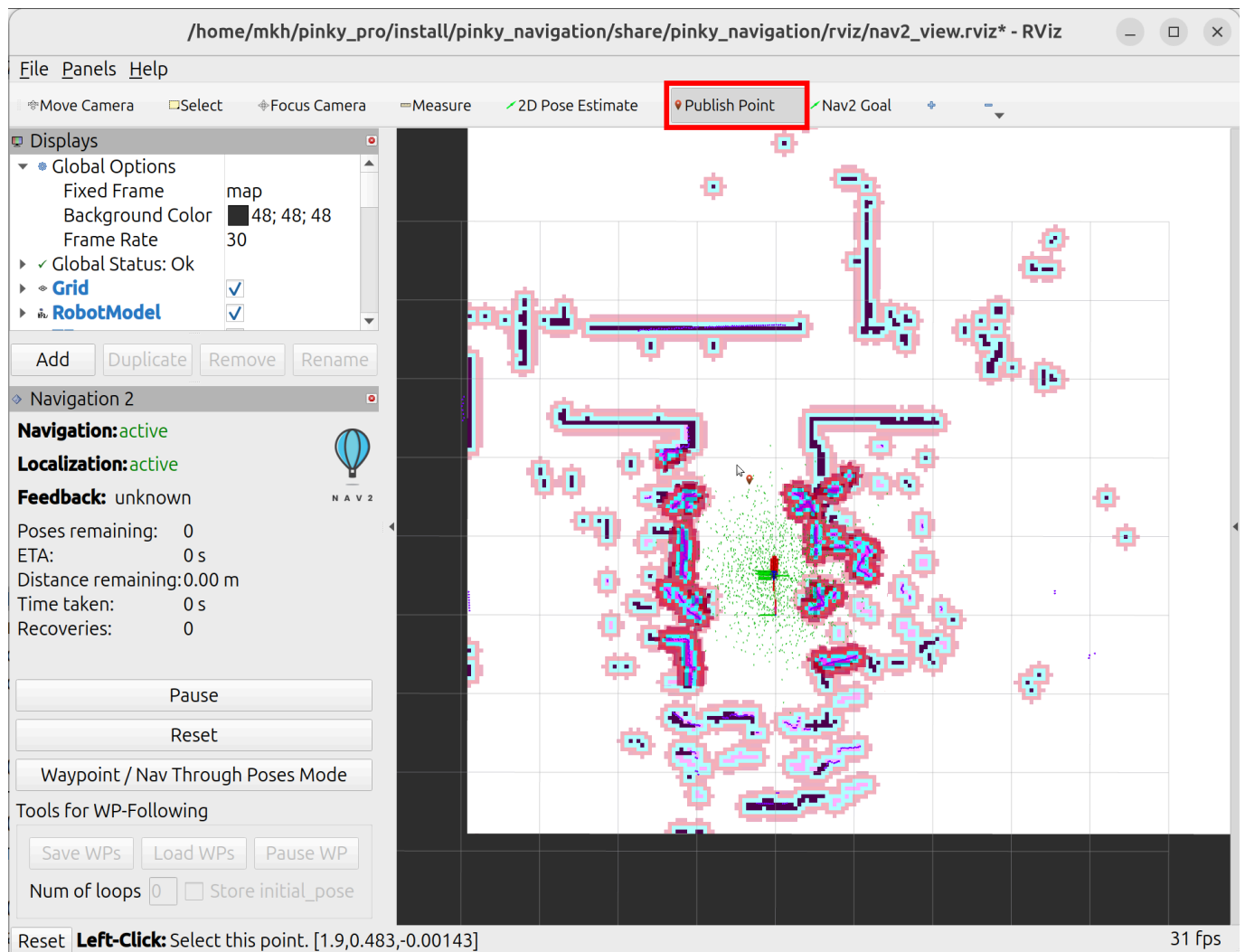
Goal succeeded!
```

여러 좌표 보내기

먼저 터미널을 새로 열어서 /clicked_point 토픽을 구독

```
pl3@desktop:~$ ros2 topic echo /clicked_point
```

Publish Point를 클릭해서 보내고 싶은 위치 여러개 찍기



터미널을 확인해보면 맵위 좌표들이 나와있다

- z 좌표는 무시

```
p13@desktop:~$ ros2 topic echo /clicked_point
```

```
header:
```

```
stamp:
```

```
sec: 1753182652
```

```
nanosec: 901790314
```

```
frame_id: map
```

```
point:
```

```
x: 1.318017601966858
```

```
y: 0.9828795790672302
```

```
z: 0.0025634765625
```

```
---
```

```
header:
```

```
stamp:
```

```
sec: 1753182659
```

```
nanosec: 298633112
```

```
frame_id: map
```

```
point:
```

```
x: -1.3463728427886963
```

```
y: 0.9966180920600891
```

```
z: -0.001434326171875
```

```
---
```

```
header:
```

```
stamp:
```

```
sec: 1753182663
```

```
nanosec: 132502013
```

```
frame_id: map
```

```
point:
```

```
x: -1.3928649425506592
```

```
y: -0.36736395955085754
```

```
z: 0.002471923828125
```

```
---
```

```
□
```

waypoint list 만들기

```

[33]: goal_pose_list = []

goal_yaw1 = 0
q1 = get_quaternion_from_yaw(goal_yaw)

goal_pose1 = PoseStamped()
goal_pose1.header.frame_id = 'map'
goal_pose1.header.stamp = nav.get_clock().now().to_msg()
goal_pose1.pose.position.x = 1.31
goal_pose1.pose.position.y = 1.0
goal_pose1.pose.orientation.x = q[0]
goal_pose1.pose.orientation.y = q[1]
goal_pose1.pose.orientation.z = q[2]
goal_pose1.pose.orientation.w = q[3]
goal_pose_list.append(goal_pose1)

goal_yaw2 = 180
q2 = get_quaternion_from_yaw(goal_yaw2)

goal_pose2 = PoseStamped()
goal_pose2.header.frame_id = 'map'
goal_pose2.header.stamp = nav.get_clock().now().to_msg()
goal_pose2.pose.position.x = -1.34
goal_pose2.pose.position.y = 1.0
goal_pose2.pose.orientation.x = q2[0]
goal_pose2.pose.orientation.y = q2[1]
goal_pose2.pose.orientation.z = q2[2]
goal_pose2.pose.orientation.w = q2[3]
goal_pose_list.append(goal_pose2)

goal_yaw3 = 45
q3 = get_quaternion_from_yaw(goal_yaw3)

goal_pose3 = PoseStamped()
goal_pose3.header.frame_id = 'map'
goal_pose3.header.stamp = nav.get_clock().now().to_msg()
goal_pose3.pose.position.x = -1.39
goal_pose3.pose.position.y = -0.36
goal_pose3.pose.orientation.x = q3[0]
goal_pose3.pose.orientation.y = q3[1]
goal_pose3.pose.orientation.z = q3[2]
goal_pose3.pose.orientation.w = q3[3]
goal_pose_list.append(goal_pose3)

```

waypoint 주행 시작 후 피드백 확인해보기

```
[42]: nav_start = nav.get_clock().now()
      nav.followWaypoints(goal_pose_list)
```

```
[INFO] [1753184765.193130417] [basic_navigator]: Following 3 goals....
```

```
[42]: True
```

```
[*]: from rclpy.duration import Duration

while not nav.isTaskComplete():
    clear_output(wait=True)
    rclpy.spin_once(sub_amcl, timeout_sec=1.0)

    feedback = nav.getFeedback()

    print("===")
    print(
        'Executing current waypoint: '
        + str(feedback.current_waypoint + 1)
        + '/'
        + str(len(goal_pose_list))
    )
    now = nav.get_clock().now()

    if now - nav_start > Duration(seconds=300.0):
        navigator.cancelTask()
```

```
===
```

```
geometry_msgs.msg.Point(x=0.15460340126386996, y=0.6553634922817777, z=0.0)
```

```
geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.4807356067183247, w=0.8768655977019307)
```

```
===
```

```
Executing current waypoint: 1/3
```

nav2 와 노드 종료

```
[44]: nav.lifecycleShutdown()
      rclpy.shutdown()
```

```
[INFO] [1753184918.926622521] [basic_navigator]: Shutting down lifecycle nodes based on lifecycle_manager.
```

```
[INFO] [1753184918.927842876] [basic_navigator]: Shutting down /lifecycle_manager_localization/manager_nodes
```

```
[INFO] [1753184919.344431134] [basic_navigator]: Shutting down /lifecycle_manager_navigation/manager_nodes
```




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